Canadian Natural

Canadian Natural Resources Ltd.

Wapiti (NAD27, Zone 11N)
Padsite: 3-7-65-8W6M
CNRL Hz Wapiti 4-31-64-8W6M
Job# 25-D0914

Plan: Proposal 4

100/04-31-064-08W6/00



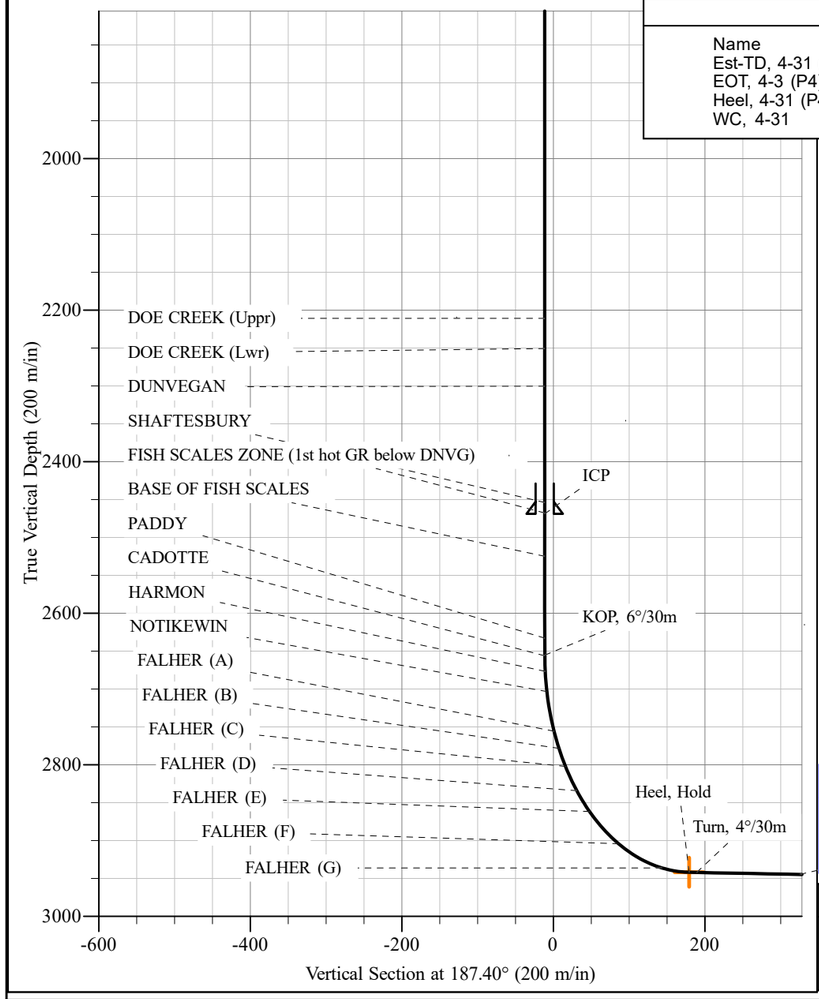
Canadian Natural

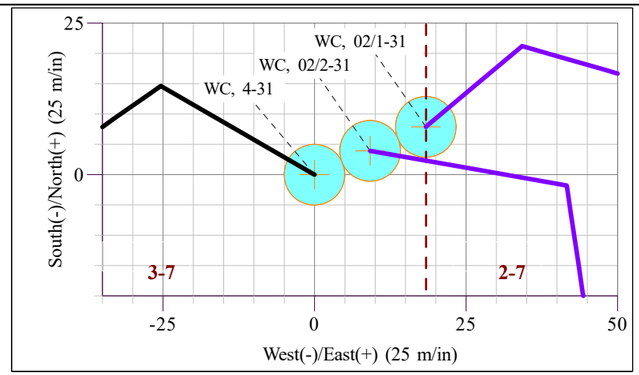
Project: Wapiti (NAD27, Zone 11N)
Site: Padsite: 3-7-65-8W6M
Well: CNRL Hz Wapiti 4-31-64-8W6M
Wellbore: Job# 25-D0914
Plan: Proposal 4
Land Survey: Compass Geomatics Ltd. 06/05/24, Rev 3

PACESETTER
YOUR PARTNERS IN DRILLING PERFORMANCE

DESIGN TARGET DETAILS

Name	TVD	+N/-S	+E/-W	Northing	Easting
Est-TD, 4-31 (P3)	2989.89	-3481.88	-452.37	6049477.66	357656.82
EOT, 4-3 (P4)	2948.44	-507.90	-452.23	6052449.73	357749.83
Heel, 4-31 (P4)	2941.99	-147.39	-256.74	6052803.90	357956.45
WC, 4-31	0.00	0.00	0.00	6052943.18	358217.63





PLEASE NOTE:

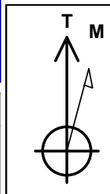
All section lines, oil/gas targets, and hard boundaries are estimates only based on the information provided on the legal land survey. These boundaries should not be taken as absolute and Pacesetter Directional Drilling Ltd. assumes no liability for the accuracy of their representation.

150m Hard Boundary (HB)

From WC: Est Enter 278.2m S, Est Exit 428.2m S
From WC: West Edge is 635.9m W
West HB is along Sec 6 Only (Not Sec 31)

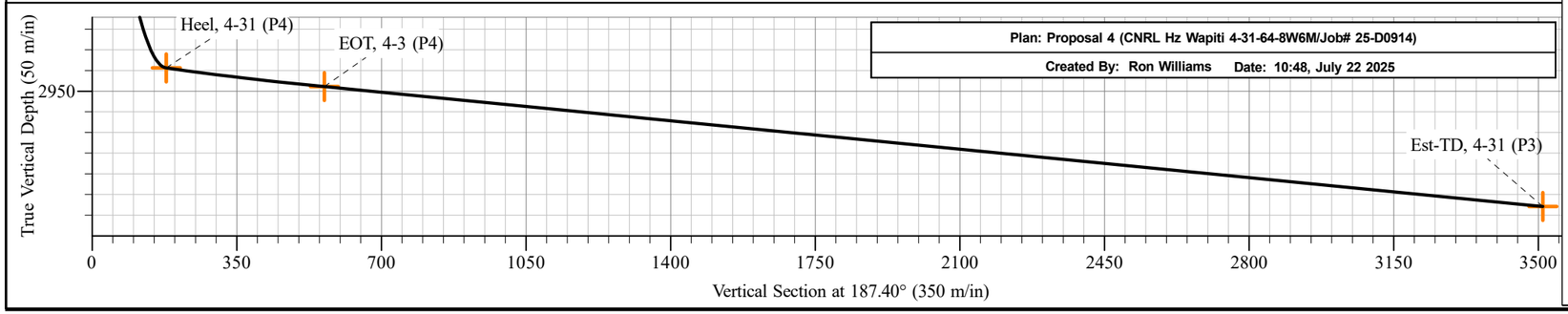
PROPOSED COORDINATES:

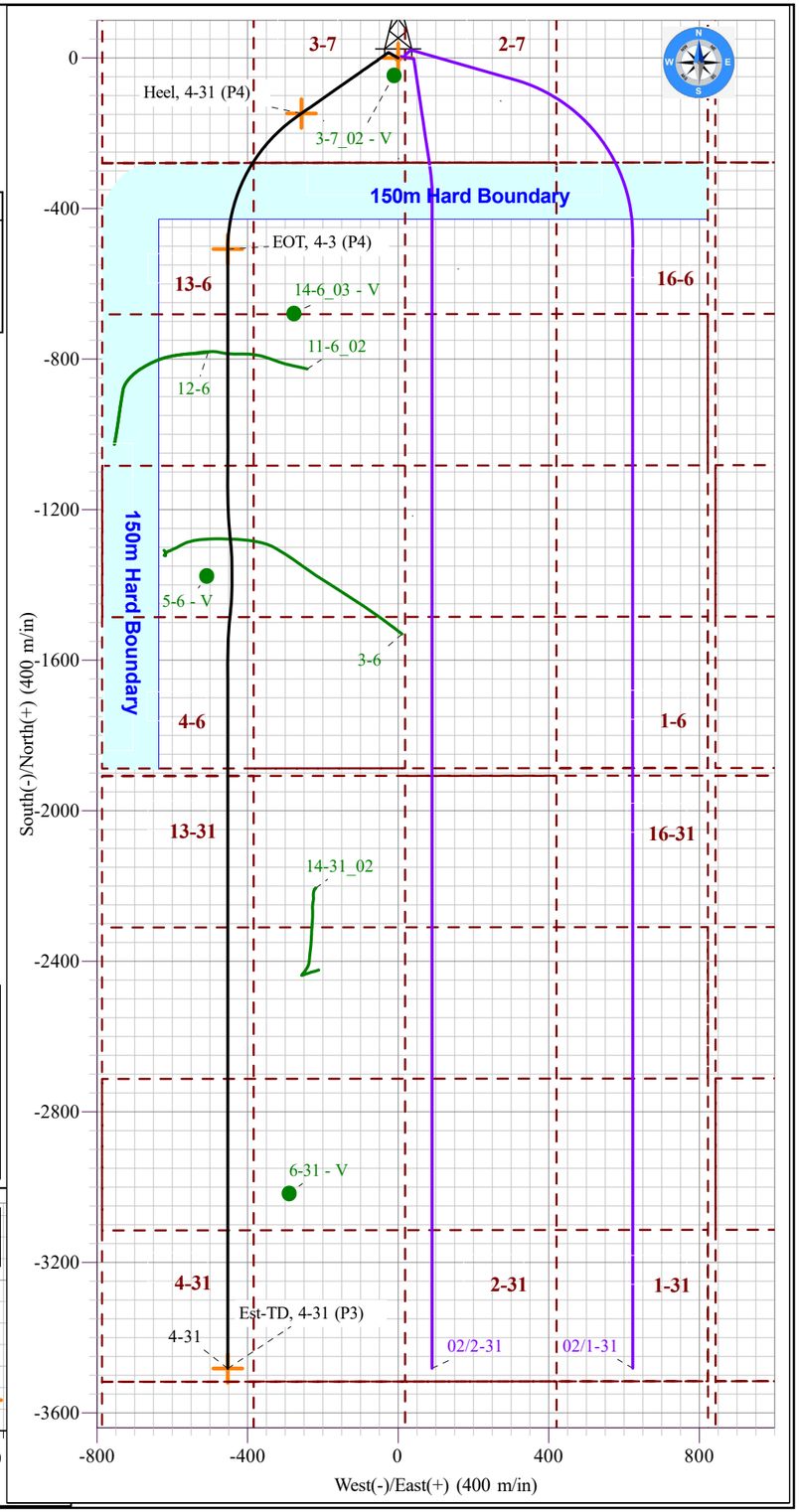
WC: 278.2m N & 785.9m E of Sec. 7
TD:



Azimuths to True North
Magnetic North: 14.70°

Magnetic Field
Strength: 55829.9nT
Dip Angle: 74.42°
Date: 2025-08-01
Model: HDGM





Database:	Compass	**Local Co-ordinate Reference:**	Well CNRL Hz Wapiti 4-31-64-8W6M
Company:	Canadian Natural Resources Ltd.	**TVD Reference:**	Est-KB @ 1116.69m
Project:	Wapiti (NAD27, Zone 11N)	**MD Reference:**	Est-KB @ 1116.69m
Site:	Padsite: 3-7-65-8W6M	**North Reference:**	True
Well:	CNRL Hz Wapiti 4-31-64-8W6M	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Job# 25-D0914		
Design:	Proposal 4		

Project Wapiti (NAD27, Zone 11N)

Map System:	Universal Transverse Mercator	**System Datum:**	Mean Sea Level
Geo Datum:	NAD 1927 - Canada - Alb/BC		
Map Zone:	Zone 11N (120 W to 114 W)		Using geodetic scale factor

Well CNRL Hz Wapiti 4-31-64-8W6M

Well Position	**+N/-S**	0.00 m	**Northing:**	6,052,943.18 m	**Latitude:**	54.605960
	+E/-W	0.00 m	**Easting:**	358,217.63 m	**Longitude:**	-119.195079
Position Uncertainty		0.00 m	**Wellhead Elevation:**	m	**Ground Level:**	1,108.59 m
					Grid Convergence:	-1.79 °

Wellbore Job# 25-D0914

Magnetics

Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
HDGM	2025-08-01	14.70	74.42	55,829.9

IF SAMPLE DATE IS NOT WITHIN 3 MONTHS OF DRILLING DATE, PLEASE CONTACT THE PACESETTER OFFICE FOR UPDATED VALUES

Design Proposal 4

Audit Notes: 21-Jul-25, Add Nudge Outside SCP and in Hz leg to get SF>1 vs offsets, 22-Jul-25

Version: **Phase:** PLAN **Tie On Depth:** 0.00

Vertical Section:

Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
0.00	0.00	0.00	187.40

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Plan Sections

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	TFO (°)	Target
0.00	0.00	0.00	0.00	0.00	0.00	0.000	0.000	0.000	0.00	
615.00	0.00	0.00	615.00	0.00	0.00	0.000	0.000	0.000	0.00	
665.00	0.00	0.00	665.00	0.00	0.00	0.000	0.000	0.000	0.00	
805.00	7.00	300.00	804.65	4.27	-7.40	1.500	1.500	0.000	300.00	
905.00	7.00	300.00	903.91	10.36	-17.95	0.000	0.000	0.000	0.00	
1,045.00	0.00	0.00	1,043.56	14.63	-25.35	1.500	-1.500	0.000	180.00	
2,470.33	0.00	0.00	2,468.89	14.63	-25.35	0.000	0.000	0.000	0.00	
2,500.33	0.00	0.00	2,498.89	14.63	-25.35	0.000	0.000	0.000	0.00	
2,656.98	0.00	0.00	2,655.54	14.63	-25.35	0.000	0.000	0.000	0.00	
3,102.98	89.20	235.00	2,941.99	-147.39	-256.74	6.000	6.000	0.000	235.00	
3,117.98	89.20	235.00	2,942.20	-155.99	-269.03	0.000	0.000	0.000	0.00	
3,530.42	89.20	180.00	2,948.44	-507.90	-452.23	4.000	0.000	-4.000	-90.41	
4,200.00	89.20	180.00	2,957.77	-1,177.42	-452.26	0.000	0.000	0.000	0.00	
4,250.02	89.20	175.00	2,958.47	-1,227.37	-450.08	3.000	-0.001	-3.000	-90.05	
4,325.02	89.20	175.00	2,959.52	-1,302.08	-443.54	0.000	0.000	0.000	0.00	
4,375.02	89.20	180.00	2,960.21	-1,352.00	-441.36	3.000	0.000	3.000	90.03	
4,425.02	89.20	180.00	2,960.91	-1,402.00	-441.36	0.000	0.000	0.000	0.00	
4,475.01	89.20	185.00	2,961.61	-1,451.93	-443.54	3.000	0.000	3.000	90.03	
4,550.01	89.20	185.00	2,962.66	-1,526.63	-450.08	0.000	0.000	0.000	0.00	
4,599.97	89.20	180.00	2,963.36	-1,576.53	-452.26	3.000	0.001	-3.000	-90.01	
6,505.51	89.20	180.00	2,989.89	-3,481.88	-452.37	0.000	0.000	0.000	0.00	Est-TD, 4-31 (P3)

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Well:	CNRL Hz Wapiti 4-31-64-8W6M	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Job# 25-D0914		
Design:	Proposal 4		

Planned Survey

All annotations and formations are for the line below

Measured Depth (m)	Inclination (°)	Azimuth (°)	TVD (m)	Sub Sea Elevation (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Build Rate (°/30m)	Turn Rate (°/30m)	Dogleg Rate (°/30m)
0.00	0.00	0.00	0.00	1,116.69	0.00	0.00	0.00	0.00	0.00	0.000
Surface Casing										
615.00	0.00	0.00	615.00	501.69	0.00	0.00	0.00	0.00	0.00	0.000
Nudge, 1.5°/30m										
665.00	0.00	0.00	665.00	451.69	0.00	0.00	0.00	0.00	0.00	0.000
690.00	1.25	300.00	690.00	426.69	0.14	-0.24	-0.10	1.50	0.00	1.500
720.00	2.75	300.00	719.98	396.71	0.66	-1.14	-0.51	1.50	0.00	1.500
PASKAPOO										
735.73	3.54	300.00	735.69	381.00	1.09	-1.89	-0.84	1.50	0.00	1.500
750.00	4.25	300.00	749.92	366.77	1.58	-2.73	-1.21	1.50	0.00	1.500
780.00	5.75	300.00	779.81	336.88	2.88	-4.99	-2.22	1.50	0.00	1.500
Hold										
805.00	7.00	300.00	804.65	312.04	4.27	-7.40	-3.28	1.50	0.00	1.500
810.00	7.00	300.00	809.61	307.08	4.58	-7.92	-3.52	0.00	0.00	0.000
840.00	7.00	300.00	839.39	277.30	6.40	-11.09	-4.92	0.00	0.00	0.000
870.00	7.00	300.00	869.17	247.52	8.23	-14.26	-6.33	0.00	0.00	0.000
SCOLLARD										
896.72	7.00	300.00	895.69	221.00	9.86	-17.08	-7.58	0.00	0.00	0.000
900.00	7.00	300.00	898.94	217.75	10.06	-17.42	-7.73	0.00	0.00	0.000
Drop, 1.5°/30m										
905.00	7.00	300.00	903.91	212.78	10.36	-17.95	-7.97	0.00	0.00	0.000
930.00	5.75	300.00	928.75	187.94	11.75	-20.36	-9.03	-1.50	0.00	1.500
BEARPAW										
958.05	4.35	300.00	956.69	160.00	12.99	-22.49	-9.98	-1.50	0.00	1.500

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960.00	4.25	300.00	958.64	158.05	13.06	-22.62	-10.04	-1.50	0.00	1.500
WAPITI										
963.06	4.10	300.00	961.69	155.00	13.17	-22.81	-10.12	-1.50	0.00	1.500
990.00	2.75	300.00	988.58	128.11	13.98	-24.21	-10.74	-1.50	0.00	1.500
1,020.00	1.25	300.00	1,018.56	98.13	14.50	-25.11	-11.14	-1.50	0.00	1.500
Vertical, Hold										
1,045.00	0.00	0.00	1,043.56	73.13	14.63	-25.35	-11.25	-1.50	0.00	1.500
BELLY RIVER GROUP										
1,055.13	0.00	0.00	1,053.69	63.00	14.63	-25.35	-11.25	0.00	0.00	0.000
PUSKWASKAU										
1,443.73	0.00	0.00	1,442.29	-325.60	14.63	-25.35	-11.25	0.00	0.00	0.000
CHINOOK SAND										
1,467.43	0.00	0.00	1,465.99	-349.30	14.63	-25.35	-11.25	0.00	0.00	0.000
COLORADO										
1,501.33	0.00	0.00	1,499.89	-383.20	14.63	-25.35	-11.25	0.00	0.00	0.000
BADHEART										
1,654.83	0.00	0.00	1,653.39	-536.70	14.63	-25.35	-11.25	0.00	0.00	0.000
MUSKIKI										
1,688.43	0.00	0.00	1,686.99	-570.30	14.63	-25.35	-11.25	0.00	0.00	0.000
CARDIUM ZONE										
1,725.33	0.00	0.00	1,723.89	-607.20	14.63	-25.35	-11.25	0.00	0.00	0.000
CARDIUM SAND										
1,751.53	0.00	0.00	1,750.09	-633.40	14.63	-25.35	-11.25	0.00	0.00	0.000
KASKAPAU										
1,787.43	0.00	0.00	1,785.99	-669.30	14.63	-25.35	-11.25	0.00	0.00	0.000

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Wellbore:	Job# 25-D0914		
Design:	Proposal 4		

Planned Survey

All annotations and formations are for the line below

Measured Depth (m)	Inclination (°)	Azimuth (°)	TVD (m)	Sub Sea Elevation (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Build Rate (°/30m)	Turn Rate (°/30m)	Dogleg Rate (°/30m)
DOE CREEK (Uppr)										
2,212.43	0.00	0.00	2,210.99	-1,094.30	14.63	-25.35	-11.25	0.00	0.00	0.000
DOE CREEK (Lwr)										
2,252.23	0.00	0.00	2,250.79	-1,134.10	14.63	-25.35	-11.25	0.00	0.00	0.000
DUNVEGAN										
2,301.93	0.00	0.00	2,300.49	-1,183.80	14.63	-25.35	-11.25	0.00	0.00	0.000
SHAFTESBURY										
2,455.33	0.00	0.00	2,453.89	-1,337.20	14.63	-25.35	-11.25	0.00	0.00	0.000
FISH SCALES ZONE (1st hot GR below DNVG)										
2,469.13	0.00	0.00	2,467.69	-1,351.00	14.63	-25.35	-11.25	0.00	0.00	0.000
ICP										
2,470.33	0.00	0.00	2,468.89	-1,352.20	14.63	-25.35	-11.25	0.00	0.00	0.000
2,500.33	0.00	0.00	2,498.89	-1,382.20	14.63	-25.35	-11.25	0.00	0.00	0.000
BASE OF FISH SCALES										
2,526.43	0.00	0.00	2,524.99	-1,408.30	14.63	-25.35	-11.25	0.00	0.00	0.000
PADDY										
2,634.43	0.00	0.00	2,632.99	-1,516.30	14.63	-25.35	-11.25	0.00	0.00	0.000
KOP, 6°/30m										
2,656.98	0.00	0.00	2,655.54	-1,538.85	14.63	-25.35	-11.25	0.00	0.00	0.000
CADOTTE										
2,657.83	0.17	235.00	2,656.39	-1,539.70	14.63	-25.35	-11.25	5.99	0.00	5.990
2,670.00	2.60	235.00	2,668.55	-1,551.86	14.47	-25.59	-11.05	6.00	0.00	6.000
HARMON										
2,678.15	4.23	235.00	2,676.69	-1,560.00	14.19	-25.99	-10.72	6.00	0.00	6.000
2,700.00	8.60	235.00	2,698.40	-1,581.71	12.79	-27.99	-9.07	6.00	0.00	6.000

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NOTIKEWIN										
2,705.16	9.64	235.00	2,703.49	-1,586.80	12.32	-28.66	-8.52	6.00	0.00	6.000
2,730.00	14.60	235.00	2,727.77	-1,611.08	9.33	-32.93	-5.01	6.00	0.00	6.000
FALHER (A)										
2,759.40	20.48	235.00	2,755.79	-1,639.10	4.25	-40.19	0.97	6.00	0.00	6.000
2,760.00	20.60	235.00	2,756.35	-1,639.66	4.12	-40.36	1.11	6.00	0.00	6.000
FALHER (B)										
2,784.17	25.44	235.00	2,778.59	-1,661.90	-1.29	-48.10	7.48	6.00	0.00	6.000
2,790.00	26.60	235.00	2,783.83	-1,667.14	-2.76	-50.19	9.21	6.00	0.00	6.000
FALHER (C)										
2,811.05	30.81	235.00	2,802.29	-1,685.60	-8.56	-58.48	16.02	6.00	0.00	6.000
2,820.00	32.60	235.00	2,809.90	-1,693.21	-11.26	-62.33	19.20	6.00	0.00	6.000
2,850.00	38.60	235.00	2,834.28	-1,717.59	-21.27	-76.63	30.97	6.00	0.00	6.000
FALHER (D)										
2,850.14	38.63	235.00	2,834.39	-1,717.70	-21.32	-76.70	31.03	6.00	0.00	6.000
2,880.00	44.60	235.00	2,856.71	-1,740.02	-32.69	-92.94	44.39	6.00	0.00	6.000
FALHER (E)										
2,887.09	46.02	235.00	2,861.69	-1,745.00	-35.58	-97.07	47.79	6.00	0.00	6.000
2,910.00	50.60	235.00	2,876.92	-1,760.23	-45.39	-111.08	59.33	6.00	0.00	6.000
2,940.00	56.60	235.00	2,894.72	-1,778.03	-59.24	-130.85	75.60	6.00	0.00	6.000
FALHER (F)										
2,958.49	60.30	235.00	2,904.39	-1,787.70	-68.28	-143.76	86.23	6.00	0.00	6.000
2,970.00	62.60	235.00	2,909.89	-1,793.20	-74.07	-152.04	93.04	6.00	0.00	6.000
3,000.00	68.60	235.00	2,922.27	-1,805.58	-89.74	-174.41	111.46	6.00	0.00	6.000
3,030.00	74.60	235.00	2,931.74	-1,815.05	-106.06	-197.72	130.65	6.00	0.00	6.000

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Wellbore:	Job# 25-D0914		
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Measured Depth (m)	Inclination (°)	Azimuth (°)	TVD (m)	Sub Sea Elevation (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Build Rate (°/30m)	Turn Rate (°/30m)	Dogleg Rate (°/30m)
FALHER (G)										
3,049.60	78.52	235.00	2,936.29	-1,819.60	-116.99	-213.33	143.50	6.00	0.00	6.000
3,060.00	80.60	235.00	2,938.17	-1,821.48	-122.86	-221.71	150.40	6.00	0.00	6.000
3,090.00	86.60	235.00	2,941.51	-1,824.82	-139.95	-246.12	170.49	6.00	0.00	6.000
Heel, Hold										
3,102.98	89.20	235.00	2,941.99	-1,825.30	-147.39	-256.74	179.24	6.00	0.00	6.000
Turn, 4°/30m										
3,117.98	89.20	235.00	2,942.20	-1,825.51	-155.99	-269.03	189.35	0.00	0.00	0.000
3,120.00	89.20	234.73	2,942.23	-1,825.54	-157.15	-270.68	190.72	-0.03	-4.00	4.000
3,150.00	89.17	230.73	2,942.65	-1,825.96	-175.31	-294.55	211.80	-0.03	-4.00	4.000
3,180.00	89.15	226.73	2,943.09	-1,826.40	-195.10	-317.09	234.32	-0.02	-4.00	4.000
3,210.00	89.13	222.73	2,943.55	-1,826.86	-216.40	-338.19	258.17	-0.02	-4.00	4.000
3,240.00	89.12	218.73	2,944.00	-1,827.31	-239.13	-357.76	283.23	-0.01	-4.00	4.000
3,270.00	89.11	214.73	2,944.47	-1,827.78	-263.17	-375.70	309.38	-0.01	-4.00	4.000
Est Entry 150m Hard Boundary										
3,288.04	89.10	212.32	2,944.75	-1,828.06	-278.20	-385.66	325.57	-0.01	-4.00	4.000
3,300.00	89.10	210.73	2,944.94	-1,828.25	-288.40	-391.91	336.48	0.00	-4.00	4.000
3,330.00	89.10	206.73	2,945.41	-1,828.72	-314.69	-406.33	364.42	0.00	-4.00	4.000
3,360.00	89.10	202.73	2,945.88	-1,829.19	-341.93	-418.87	393.05	0.00	-4.00	4.000
3,390.00	89.11	198.73	2,946.35	-1,829.66	-369.98	-429.48	422.23	0.01	-4.00	4.000
3,420.00	89.12	194.73	2,946.81	-1,830.12	-398.70	-438.12	451.83	0.01	-4.00	4.000
3,450.00	89.14	190.73	2,947.27	-1,830.58	-427.96	-444.72	481.69	0.02	-4.00	4.000
Est Exit 150m Hard Boundary										
3,450.25	89.14	190.69	2,947.27	-1,830.58	-428.20	-444.77	481.94	0.02	-4.00	4.000
3,480.00	89.16	186.73	2,947.72	-1,831.03	-457.60	-449.27	511.67	0.02	-4.00	4.000

Database:	Compass	**Local Co-ordinate Reference:**	Well CNRL Hz Wapiti 4-31-64-8W6M
Company:	Canadian Natural Resources Ltd.	**TVD Reference:**	Est-KB @ 1116.69m
Project:	Wapiti (NAD27, Zone 11N)	**MD Reference:**	Est-KB @ 1116.69m
Site:	Padsite: 3-7-65-8W6M	**North Reference:**	True
Well:	CNRL Hz Wapiti 4-31-64-8W6M	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Job# 25-D0914		
Design:	Proposal 4		

Planned Survey

All annotations and formations are for the line below

Measured Depth (m)	Inclination (°)	Azimuth (°)	TVD (m)	Sub Sea Elevation (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Build Rate (°/30m)	Turn Rate (°/30m)	Dogleg Rate (°/30m)
3,510.00	89.18	182.73	2,948.15	-1,831.46	-487.49	-451.74	541.63	0.02	-4.00	4.000
EOT, Hold										
3,530.42	89.20	180.00	2,948.44	-1,831.75	-507.90	-452.23	561.93	0.03	-4.00	3.999
3,540.00	89.20	180.00	2,948.57	-1,831.88	-517.48	-452.23	571.43	0.00	0.00	0.000
3,570.00	89.20	180.00	2,948.99	-1,832.30	-547.48	-452.23	601.18	0.00	0.00	0.000
3,600.00	89.20	180.00	2,949.41	-1,832.72	-577.47	-452.23	630.93	0.00	0.00	0.000
3,630.00	89.20	180.00	2,949.83	-1,833.14	-607.47	-452.23	660.67	0.00	0.00	0.000
3,660.00	89.20	180.00	2,950.25	-1,833.56	-637.47	-452.23	690.42	0.00	0.00	0.000
3,690.00	89.20	180.00	2,950.66	-1,833.97	-667.47	-452.24	720.17	0.00	0.00	0.000
3,720.00	89.20	180.00	2,951.08	-1,834.39	-697.46	-452.24	749.92	0.00	0.00	0.000
3,750.00	89.20	180.00	2,951.50	-1,834.81	-727.46	-452.24	779.66	0.00	0.00	0.000
3,780.00	89.20	180.00	2,951.92	-1,835.23	-757.46	-452.24	809.41	0.00	0.00	0.000
3,810.00	89.20	180.00	2,952.34	-1,835.65	-787.45	-452.24	839.16	0.00	0.00	0.000
3,840.00	89.20	180.00	2,952.75	-1,836.06	-817.45	-452.24	868.90	0.00	0.00	0.000
3,870.00	89.20	180.00	2,953.17	-1,836.48	-847.45	-452.24	898.65	0.00	0.00	0.000
3,900.00	89.20	180.00	2,953.59	-1,836.90	-877.45	-452.25	928.40	0.00	0.00	0.000
3,930.00	89.20	180.00	2,954.01	-1,837.32	-907.44	-452.25	958.15	0.00	0.00	0.000
3,960.00	89.20	180.00	2,954.43	-1,837.74	-937.44	-452.25	987.89	0.00	0.00	0.000
3,990.00	89.20	180.00	2,954.84	-1,838.15	-967.44	-452.25	1,017.64	0.00	0.00	0.000
4,020.00	89.20	180.00	2,955.26	-1,838.57	-997.43	-452.25	1,047.39	0.00	0.00	0.000
4,050.00	89.20	180.00	2,955.68	-1,838.99	-1,027.43	-452.25	1,077.14	0.00	0.00	0.000
4,080.00	89.20	180.00	2,956.10	-1,839.41	-1,057.43	-452.25	1,106.88	0.00	0.00	0.000
4,110.00	89.20	180.00	2,956.52	-1,839.83	-1,087.42	-452.26	1,136.63	0.00	0.00	0.000
4,140.00	89.20	180.00	2,956.93	-1,840.24	-1,117.42	-452.26	1,166.38	0.00	0.00	0.000
4,170.00	89.20	180.00	2,957.35	-1,840.66	-1,147.42	-452.26	1,196.12	0.00	0.00	0.000

Database:	Compass	**Local Co-ordinate Reference:**	Well CNRL Hz Wapiti 4-31-64-8W6M
Company:	Canadian Natural Resources Ltd.	**TVD Reference:**	Est-KB @ 1116.69m
Project:	Wapiti (NAD27, Zone 11N)	**MD Reference:**	Est-KB @ 1116.69m
Site:	Padsite: 3-7-65-8W6M	**North Reference:**	True
Well:	CNRL Hz Wapiti 4-31-64-8W6M	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Job# 25-D0914		
Design:	Proposal 4		

Planned Survey

All annotations and formations are for the line below

Measured Depth (m)	Inclination (°)	Azimuth (°)	TVD (m)	Sub Sea Elevation (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Build Rate (°/30m)	Turn Rate (°/30m)	Dogleg Rate (°/30m)
3°/30m										
4,200.00	89.20	180.00	2,957.77	-1,841.08	-1,177.42	-452.26	1,225.87	0.00	0.00	0.000
4,230.00	89.20	177.00	2,958.19	-1,841.50	-1,207.40	-451.48	1,255.50	0.00	-3.00	3.000
Hold										
4,250.02	89.20	175.00	2,958.47	-1,841.78	-1,227.37	-450.08	1,275.13	0.00	-3.00	3.000
4,260.00	89.20	175.00	2,958.61	-1,841.92	-1,237.31	-449.21	1,284.87	0.00	0.00	0.001
4,290.00	89.20	175.00	2,959.03	-1,842.34	-1,267.19	-446.60	1,314.17	0.00	0.00	0.000
4,320.00	89.20	175.00	2,959.45	-1,842.76	-1,297.07	-443.98	1,343.47	0.00	0.00	0.000
3°/30m										
4,325.02	89.20	175.00	2,959.52	-1,842.83	-1,302.08	-443.54	1,348.37	0.00	0.00	0.000
4,350.00	89.20	177.50	2,959.87	-1,843.18	-1,327.00	-441.91	1,372.87	0.00	3.00	3.000
Hold										
4,375.02	89.20	180.00	2,960.21	-1,843.52	-1,352.01	-441.36	1,397.60	0.00	3.00	3.000
4,380.00	89.20	180.00	2,960.28	-1,843.59	-1,356.99	-441.36	1,402.54	0.00	0.00	0.000
4,410.00	89.20	180.00	2,960.70	-1,844.01	-1,386.98	-441.36	1,432.29	0.00	0.00	0.000
3°/30m										
4,425.02	89.20	180.00	2,960.91	-1,844.22	-1,402.00	-441.36	1,447.18	0.00	0.00	0.000
4,440.00	89.20	181.50	2,961.12	-1,844.43	-1,416.98	-441.56	1,462.06	0.00	3.00	3.000
4,470.00	89.20	184.50	2,961.54	-1,844.85	-1,446.93	-443.13	1,491.96	0.00	3.00	3.000
Hold										
4,475.01	89.20	185.00	2,961.61	-1,844.92	-1,451.92	-443.54	1,496.97	0.00	3.00	3.000
4,500.00	89.20	185.00	2,961.96	-1,845.27	-1,476.82	-445.72	1,521.93	0.00	0.00	0.000
4,530.00	89.20	185.00	2,962.38	-1,845.69	-1,506.70	-448.34	1,551.90	0.00	0.00	0.000
3°/30m										
4,550.01	89.20	185.00	2,962.66	-1,845.97	-1,526.63	-450.08	1,571.90	0.00	0.00	0.000

Database:	Compass	**Local Co-ordinate Reference:**	Well CNRL Hz Wapiti 4-31-64-8W6M
Company:	Canadian Natural Resources Ltd.	**TVD Reference:**	Est-KB @ 1116.69m
Project:	Wapiti (NAD27, Zone 11N)	**MD Reference:**	Est-KB @ 1116.69m
Site:	Padsite: 3-7-65-8W6M	**North Reference:**	True
Well:	CNRL Hz Wapiti 4-31-64-8W6M	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Job# 25-D0914		
Design:	Proposal 4		

Planned Survey

All annotations and formations are for the line below

Measured Depth (m)	Inclination (°)	Azimuth (°)	TVD (m)	Sub Sea Elevation (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Build Rate (°/30m)	Turn Rate (°/30m)	Dogleg Rate (°/30m)
4,560.00	89.20	184.00	2,962.80	-1,846.11	-1,536.59	-450.86	1,581.87	0.00	-3.00	2.999
4,590.00	89.20	181.00	2,963.22	-1,846.53	-1,566.55	-452.17	1,611.76	0.00	-3.00	3.000
Hold										
4,599.97	89.20	180.00	2,963.36	-1,846.67	-1,576.52	-452.26	1,621.65	0.00	-3.00	3.000
4,620.00	89.20	180.00	2,963.63	-1,846.94	-1,596.55	-452.26	1,641.51	0.00	0.00	0.001
4,650.00	89.20	180.00	2,964.05	-1,847.36	-1,626.55	-452.26	1,671.26	0.00	0.00	0.000
4,680.00	89.20	180.00	2,964.47	-1,847.78	-1,656.54	-452.27	1,701.01	0.00	0.00	0.000
4,710.00	89.20	180.00	2,964.89	-1,848.20	-1,686.54	-452.27	1,730.76	0.00	0.00	0.000
4,740.00	89.20	180.00	2,965.31	-1,848.62	-1,716.54	-452.27	1,760.50	0.00	0.00	0.000
4,770.00	89.20	180.00	2,965.72	-1,849.03	-1,746.54	-452.27	1,790.25	0.00	0.00	0.000
4,800.00	89.20	180.00	2,966.14	-1,849.45	-1,776.53	-452.27	1,820.00	0.00	0.00	0.000
4,830.00	89.20	180.00	2,966.56	-1,849.87	-1,806.53	-452.27	1,849.74	0.00	0.00	0.000
4,860.00	89.20	180.00	2,966.98	-1,850.29	-1,836.53	-452.28	1,879.49	0.00	0.00	0.000
4,890.00	89.20	180.00	2,967.39	-1,850.70	-1,866.52	-452.28	1,909.24	0.00	0.00	0.000
4,920.00	89.20	180.00	2,967.81	-1,851.12	-1,896.52	-452.28	1,938.99	0.00	0.00	0.000
4,950.00	89.20	180.00	2,968.23	-1,851.54	-1,926.52	-452.28	1,968.73	0.00	0.00	0.000
4,980.00	89.20	180.00	2,968.65	-1,851.96	-1,956.52	-452.28	1,998.48	0.00	0.00	0.000
5,010.00	89.20	180.00	2,969.07	-1,852.38	-1,986.51	-452.28	2,028.23	0.00	0.00	0.000
5,040.00	89.20	180.00	2,969.48	-1,852.79	-2,016.51	-452.29	2,057.98	0.00	0.00	0.000
5,070.00	89.20	180.00	2,969.90	-1,853.21	-2,046.51	-452.29	2,087.72	0.00	0.00	0.000
5,100.00	89.20	180.00	2,970.32	-1,853.63	-2,076.50	-452.29	2,117.47	0.00	0.00	0.000
5,130.00	89.20	180.00	2,970.74	-1,854.05	-2,106.50	-452.29	2,147.22	0.00	0.00	0.000
5,160.00	89.20	180.00	2,971.15	-1,854.46	-2,136.50	-452.29	2,176.96	0.00	0.00	0.000
5,190.00	89.20	180.00	2,971.57	-1,854.88	-2,166.50	-452.29	2,206.71	0.00	0.00	0.000
5,220.00	89.20	180.00	2,971.99	-1,855.30	-2,196.49	-452.30	2,236.46	0.00	0.00	0.000

Database:	Compass	**Local Co-ordinate Reference:**	Well CNRL Hz Wapiti 4-31-64-8W6M
Company:	Canadian Natural Resources Ltd.	**TVD Reference:**	Est-KB @ 1116.69m
Project:	Wapiti (NAD27, Zone 11N)	**MD Reference:**	Est-KB @ 1116.69m
Site:	Padsite: 3-7-65-8W6M	**North Reference:**	True
Well:	CNRL Hz Wapiti 4-31-64-8W6M	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Job# 25-D0914		
Design:	Proposal 4		

Planned Survey

All annotations and formations are for the line below

Measured Depth (m)	Inclination (°)	Azimuth (°)	TVD (m)	Sub Sea Elevation (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Build Rate (°/30m)	Turn Rate (°/30m)	Dogleg Rate (°/30m)
5,250.00	89.20	180.00	2,972.41	-1,855.72	-2,226.49	-452.30	2,266.21	0.00	0.00	0.000
5,280.00	89.20	180.00	2,972.82	-1,856.13	-2,256.49	-452.30	2,295.95	0.00	0.00	0.000
5,310.00	89.20	180.00	2,973.24	-1,856.55	-2,286.48	-452.30	2,325.70	0.00	0.00	0.000
5,340.00	89.20	180.00	2,973.66	-1,856.97	-2,316.48	-452.30	2,355.45	0.00	0.00	0.000
5,370.00	89.20	180.00	2,974.08	-1,857.39	-2,346.48	-452.30	2,385.20	0.00	0.00	0.000
5,400.00	89.20	180.00	2,974.50	-1,857.81	-2,376.48	-452.31	2,414.94	0.00	0.00	0.000
5,430.00	89.20	180.00	2,974.91	-1,858.22	-2,406.47	-452.31	2,444.69	0.00	0.00	0.000
5,460.00	89.20	180.00	2,975.33	-1,858.64	-2,436.47	-452.31	2,474.44	0.00	0.00	0.000
5,490.00	89.20	180.00	2,975.75	-1,859.06	-2,466.47	-452.31	2,504.19	0.00	0.00	0.000
5,520.00	89.20	180.00	2,976.17	-1,859.48	-2,496.46	-452.31	2,533.93	0.00	0.00	0.000
5,550.00	89.20	180.00	2,976.58	-1,859.89	-2,526.46	-452.32	2,563.68	0.00	0.00	0.000
5,580.00	89.20	180.00	2,977.00	-1,860.31	-2,556.46	-452.32	2,593.43	0.00	0.00	0.000
5,610.00	89.20	180.00	2,977.42	-1,860.73	-2,586.45	-452.32	2,623.17	0.00	0.00	0.000
5,640.00	89.20	180.00	2,977.84	-1,861.15	-2,616.45	-452.32	2,652.92	0.00	0.00	0.000
5,670.00	89.20	180.00	2,978.26	-1,861.57	-2,646.45	-452.32	2,682.67	0.00	0.00	0.000
5,700.00	89.20	180.00	2,978.67	-1,861.98	-2,676.45	-452.32	2,712.42	0.00	0.00	0.000
5,730.00	89.20	180.00	2,979.09	-1,862.40	-2,706.44	-452.33	2,742.16	0.00	0.00	0.000
5,760.00	89.20	180.00	2,979.51	-1,862.82	-2,736.44	-452.33	2,771.91	0.00	0.00	0.000
5,790.00	89.20	180.00	2,979.93	-1,863.24	-2,766.44	-452.33	2,801.66	0.00	0.00	0.000
5,820.00	89.20	180.00	2,980.34	-1,863.65	-2,796.43	-452.33	2,831.41	0.00	0.00	0.000
5,850.00	89.20	180.00	2,980.76	-1,864.07	-2,826.43	-452.33	2,861.15	0.00	0.00	0.000
5,880.00	89.20	180.00	2,981.18	-1,864.49	-2,856.43	-452.33	2,890.90	0.00	0.00	0.000
5,910.00	89.20	180.00	2,981.60	-1,864.91	-2,886.43	-452.34	2,920.65	0.00	0.00	0.000
5,940.00	89.20	180.00	2,982.02	-1,865.33	-2,916.42	-452.34	2,950.39	0.00	0.00	0.000
5,970.00	89.20	180.00	2,982.43	-1,865.74	-2,946.42	-452.34	2,980.14	0.00	0.00	0.000

Database:	Compass	**Local Co-ordinate Reference:**	Well CNRL Hz Wapiti 4-31-64-8W6M
Company:	Canadian Natural Resources Ltd.	**TVD Reference:**	Est-KB @ 1116.69m
Project:	Wapiti (NAD27, Zone 11N)	**MD Reference:**	Est-KB @ 1116.69m
Site:	Padsite: 3-7-65-8W6M	**North Reference:**	True
Well:	CNRL Hz Wapiti 4-31-64-8W6M	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Job# 25-D0914		
Design:	Proposal 4		

Planned Survey

All annotations and formations are for the line below

Measured Depth (m)	Inclination (°)	Azimuth (°)	TVD (m)	Sub Sea Elevation (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Build Rate (°/30m)	Turn Rate (°/30m)	Dogleg Rate (°/30m)
6,000.00	89.20	180.00	2,982.85	-1,866.16	-2,976.42	-452.34	3,009.89	0.00	0.00	0.000
6,030.00	89.20	180.00	2,983.27	-1,866.58	-3,006.41	-452.34	3,039.64	0.00	0.00	0.000
6,060.00	89.20	180.00	2,983.69	-1,867.00	-3,036.41	-452.34	3,069.38	0.00	0.00	0.000
6,090.00	89.20	180.00	2,984.10	-1,867.41	-3,066.41	-452.35	3,099.13	0.00	0.00	0.000
6,120.00	89.20	180.00	2,984.52	-1,867.83	-3,096.41	-452.35	3,128.88	0.00	0.00	0.000
6,150.00	89.20	180.00	2,984.94	-1,868.25	-3,126.40	-452.35	3,158.63	0.00	0.00	0.000
6,180.00	89.20	180.00	2,985.36	-1,868.67	-3,156.40	-452.35	3,188.37	0.00	0.00	0.000
6,210.00	89.20	180.00	2,985.78	-1,869.09	-3,186.40	-452.35	3,218.12	0.00	0.00	0.000
6,240.00	89.20	180.00	2,986.19	-1,869.50	-3,216.39	-452.35	3,247.87	0.00	0.00	0.000
6,270.00	89.20	180.00	2,986.61	-1,869.92	-3,246.39	-452.36	3,277.61	0.00	0.00	0.000
6,300.00	89.20	180.00	2,987.03	-1,870.34	-3,276.39	-452.36	3,307.36	0.00	0.00	0.000
6,330.00	89.20	180.00	2,987.45	-1,870.76	-3,306.38	-452.36	3,337.11	0.00	0.00	0.000
6,360.00	89.20	180.00	2,987.86	-1,871.17	-3,336.38	-452.36	3,366.86	0.00	0.00	0.000
6,390.00	89.20	180.00	2,988.28	-1,871.59	-3,366.38	-452.36	3,396.60	0.00	0.00	0.000
6,420.00	89.20	180.00	2,988.70	-1,872.01	-3,396.38	-452.37	3,426.35	0.00	0.00	0.000
6,450.00	89.20	180.00	2,989.12	-1,872.43	-3,426.37	-452.37	3,456.10	0.00	0.00	0.000
6,480.00	89.20	180.00	2,989.53	-1,872.84	-3,456.37	-452.37	3,485.85	0.00	0.00	0.000
Est-TD										
6,505.51	89.20	180.00	2,989.89	-1,873.20	-3,481.88	-452.37	3,511.14	0.00	0.00	0.000

Casing Points

Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (mm)	Hole Diameter (mm)
2,470.33	2,468.89	ICP	139.70	152.40

Database:	Compass	**Local Co-ordinate Reference:**	Well CNRL Hz Wapiti 4-31-64-8W6M
Company:	Canadian Natural Resources Ltd.	**TVD Reference:**	Est-KB @ 1116.69m
Project:	Wapiti (NAD27, Zone 11N)	**MD Reference:**	Est-KB @ 1116.69m
Site:	Padsite: 3-7-65-8W6M	**North Reference:**	True
Well:	CNRL Hz Wapiti 4-31-64-8W6M	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Job# 25-D0914		
Design:	Proposal 4		

Formations

Measured Depth (m)	Vertical Depth (m)	Name	Lithology	Dip (°)	Dip Direction (°)
735.73	735.69	PASKAPOO			
896.72	895.69	SCOLLARD			
958.05	956.69	BEARPAW			
963.06	961.69	WAPITI			
1,055.13	1,053.69	BELLY RIVER GROUP			
1,443.73	1,442.29	PUSKWASKAU			
1,467.43	1,465.99	CHINOOK SAND			
1,501.33	1,499.89	COLORADO			
1,654.83	1,653.39	BADHEART			
1,688.43	1,686.99	MUSKIKI			
1,725.33	1,723.89	CARDIUM ZONE			
1,751.53	1,750.09	CARDIUM SAND			
1,787.43	1,785.99	KASKAPAU			
2,212.43	2,210.99	DOE CREEK (Uppr)			
2,252.23	2,250.79	DOE CREEK (Lwr)			
2,301.93	2,300.49	DUNVEGAN			
2,455.33	2,453.89	SHAFTESBURY			
2,469.13	2,467.69	FISH SCALES ZONE (1st hot GR below DNVG)			
2,526.43	2,524.99	BASE OF FISH SCALES			
2,634.43	2,632.99	PADDY			
2,657.83	2,656.39	CADOTTE			
2,678.15	2,676.69	HARMON			
2,705.16	2,703.49	NOTIKEWIN			
2,759.40	2,755.79	FALHER (A)			
2,784.17	2,778.59	FALHER (B)			
2,811.05	2,802.29	FALHER (C)			
2,850.14	2,834.39	FALHER (D)			
2,887.09	2,861.69	FALHER (E)			

Database:	Compass	**Local Co-ordinate Reference:**	Well CNRL Hz Wapiti 4-31-64-8W6M
Company:	Canadian Natural Resources Ltd.	**TVD Reference:**	Est-KB @ 1116.69m
Project:	Wapiti (NAD27, Zone 11N)	**MD Reference:**	Est-KB @ 1116.69m
Site:	Padsite: 3-7-65-8W6M	**North Reference:**	True
Well:	CNRL Hz Wapiti 4-31-64-8W6M	**Survey Calculation Method:**	Minimum Curvature
Wellbore:	Job# 25-D0914		
Design:	Proposal 4		

2,958.49	2,904.39	FALHER (F)
3,049.60	2,936.29	FALHER (G)

Plan Annotations

Measured Depth (m)	Vertical Depth (m)	Local Coordinates +N/-S (m)	Local Coordinates +E/-W (m)	Comment
615.00	615.00	0.00	0.00	Surface Casing
665.00	665.00	0.00	0.00	Nudge, 1.5°/30m
805.00	804.65	4.27	-7.40	Hold
905.00	903.91	10.36	-17.95	Drop, 1.5°/30m
1,045.00	1,043.56	14.63	-25.35	Vertical, Hold
2,470.33	2,468.89	14.63	-25.35	ICP
2,656.98	2,655.54	14.63	-25.35	KOP, 6°/30m
3,102.98	2,941.99	14.63	-25.35	Heel, Hold
3,117.98	2,942.20	-147.39	-256.74	Turn, 4°/30m
3,288.04	2,944.75	-3,481.88	-452.37	Est Entry 150m Hard Boundary
3,450.25	2,947.27	-278.20	-385.66	Est Exit 150m Hard Boundary
3,530.42	2,948.44	-155.99	-269.03	EOT, Hold
4,200.00	2,957.77	-507.90	-452.23	3°/30m
4,250.02	2,958.47	-1,177.42	-452.26	Hold
4,325.02	2,959.52	-1,227.37	-450.08	3°/30m
4,375.02	2,960.21	-1,302.08	-443.54	Hold
4,425.02	2,960.91	-1,352.00	-441.36	3°/30m
4,475.01	2,961.61	-1,402.00	-441.36	Hold
4,550.01	2,962.66	-1,451.93	-443.54	3°/30m
4,599.97	2,963.36	-1,526.63	-450.08	Hold
6,505.51	2,989.89	-1,576.53	-452.26	Est-TD